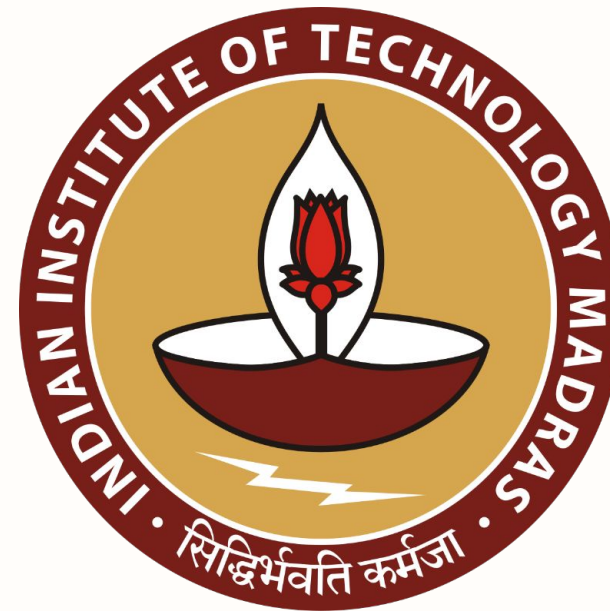


Profit Optimization Using Sales Analytic

(Capstone Project – BDM)



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Stage: Viva Voce Presentation

Problem Statement



Identified Challenges:



Blocked Inventory

- Many SKUs have zero or minimal sales
- Leads to capital blockage and storage costs



High Returns & Cancellations

- Over 10% of orders are cancelled or returned
- Results in revenue leakage and customer dissatisfaction



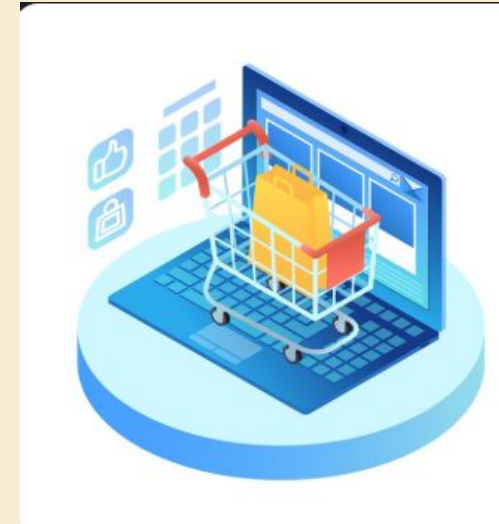
Category Concentration Risk

- Over 80% of sales come from just two product categories
- Limits diversification and increases business risk

Organization Overview

About TrendyThreads Pvt. Ltd.:

- B2C online fashion retailer based in India
- Sells women's ethnic & western wear (Kurtas, Sets, Tops)
- Operates mainly through Amazon India
- Uses both Amazon Fulfilled (FBA) and Merchant Fulfilled models
- Targets urban and semi-urban female consumers



Data Description

This dataset, collected from Amazon sales reports, offers detailed insights into order trends, SKU performance, fulfillment types, and regional sales for TrendyThreads Pvt. Ltd.

Data Overview

- **Source:** Kaggle (Original by Anil.Scraping)
- **Time Period Covered:** [31 March to 29 June 2022]
- **Total Records:** ~10,000 orders
- **Tools Used:** Google Sheets, Python (Pandas)
- **Key Fields:**
 - Order ID
 - SKU
 - Category
 - Order Status (Delivered, Returned, Cancelled)
 - Order Date
 - Fulfillment (Amazon/Merchant)
 - Sales Amount
 - Quantity Ordered
 - Customer Location (State, City)

	A	B	C	D	E	F	G	H	I	J	K
1	index	Order ID	Date	Status	Fulfillment	Sales Chanr	ship-service	Style	SKU	Category	Size
2		0 405-8078784-57	2022-04-30	Cancelled	Merchant	Amazon.in	Standard	SET389	SET389-KR-NP-Set		S
3		1 171-9198151-11	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	JNE3781	JNE3781-KR-XX kurta		3XL
4		2 404-0687676-72	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3371	JNE3371-KR-XL kurta		XL
5		3 403-9615377-81	2022-04-30	Cancelled	Merchant	Amazon.in	Standard	J0341	J0341-DR-L Western Dress		L
6		4 407-1069790-72	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3671	JNE3671-TU-XX Top		3XL
7		5 404-1490984-45	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	SET264	SET264-KR-NP-Set		XL
8		6 408-5748499-68	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	J0095	J0095-SET-L Set		L
9		7 406-7807733-37	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	JNE3405	JNE3405-KR-S kurta		S
10		9 402-4393761-03	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3461	JNE3461-KR-XX kurta		XXL
11		10 407-5633625-69	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3160	JNE3160-KR-G kurta		S
12		11 171-4638481-63	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3500	JNE3500-KR-XS kurta		XS
13		12 405-5513694-81	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	JNE3405	JNE3405-KR-XS kurta		XS
14		13 408-7955685-30	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	SET182	SET182-KR-DH-Set		XS
15		14 408-1298370-19	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	J0351	J0351-SET-L Set		L
16		15 403-4965581-95	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	PJNE3368	PJNE3368-KR-6 kurta		6XL
17		16 406-9379318-65	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3721	JNE3721-KR-XX kurta		XXL
18		17 405-9013803-80	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	JNE3405	JNE3405-KR-XL kurta		XL
19		18 402-4030358-58	2022-04-30	Shipped - Delive	Merchant	Amazon.in	Standard	JNE3697	JNE3697-KR-XX kurta		XXL
20		19 405-5957858-10	2022-04-30	Shipped	Amazon	Amazon.in	Expedited	SET254	SET254-KR-NP-Set		XS

	M	N	O	P	Q	R	S	T	U	V
1	Courier Stat	Qty	currency	Amount	ship-city	ship-state	ship-postal	ship-country	B2B	fulfilled-by
2	Unknown	0	INR	647.62	MUMBAI	MAHARASHTR	400081	IN	FALSE	Easy Ship
3	Shipped	1	INR	406	BENGALURU	KARNATAKA	560085	IN	FALSE	Easy Ship
4	Shipped	1	INR	329	NAVI MUMBAI	MAHARASHTR	410210	IN	TRUE	Amazon
5	Unknown	0	INR	753.33	PUDUCHERRY	PUDUCHERRY	605008	IN	FALSE	Easy Ship
6	Shipped	1	INR	574	CHENNAI	TAMIL NADU	600073	IN	FALSE	Amazon
7	Shipped	1	INR	824	GHAZIABAD	UTTAR PRADE	201102	IN	FALSE	Amazon
8	Shipped	1	INR	653	CHANDIGARH	CHANDIGARH	160036	IN	FALSE	Amazon
9	Shipped	1	INR	399	HYDERABAD	TELANGANA	500032	IN	FALSE	Easy Ship
10	Shipped	1	INR	363	Chennai	TAMIL NADU	600041	IN	FALSE	Amazon
11	Shipped	1	INR	685	CHENNAI	TAMIL NADU	600073	IN	FALSE	Amazon
12	Shipped	1	INR	364	NOIDA	UTTAR PRADE	201303	IN	FALSE	Amazon
13	Shipped	1	INR	399	Amravati	MAHARASHTR	444606	IN	FALSE	Easy Ship
14	Shipped	1	INR	657	MUMBAI	MAHARASHTR	400053	IN	FALSE	Amazon
15	Shipped	1	INR	771	MUMBAI	MAHARASHTR	400053	IN	FALSE	Easy Ship
16	Shipped	1	INR	544	GUNTAKAL	ANDHRA PRAD	515801	IN	FALSE	Easy Ship
17	Shipped	1	INR	329	JAIPUR	RAJASTHAN	302020	IN	FALSE	Amazon
18	Shipped	1	INR	399	NEW DELHI	DELHI	110074	IN	FALSE	Amazon
19	Shipped	1	INR	458	Gurgaon	HARYANA	122004	IN	FALSE	Easy Ship
20	Shipped	1	INR	886	BENGALURU	KARNATAKA	560017	IN	FALSE	Amazon

Methodology

Objective:

To identify patterns, inefficiencies, and opportunities within sales, inventory, and fulfillment data using structured data analysis techniques.

Approach:

Data Cleaning:

- Removed blanks, duplicates, and formatting issues using Python
- Ensured clean SKU, order status, date, and numeric fields

Descriptive Statistics & EDA:

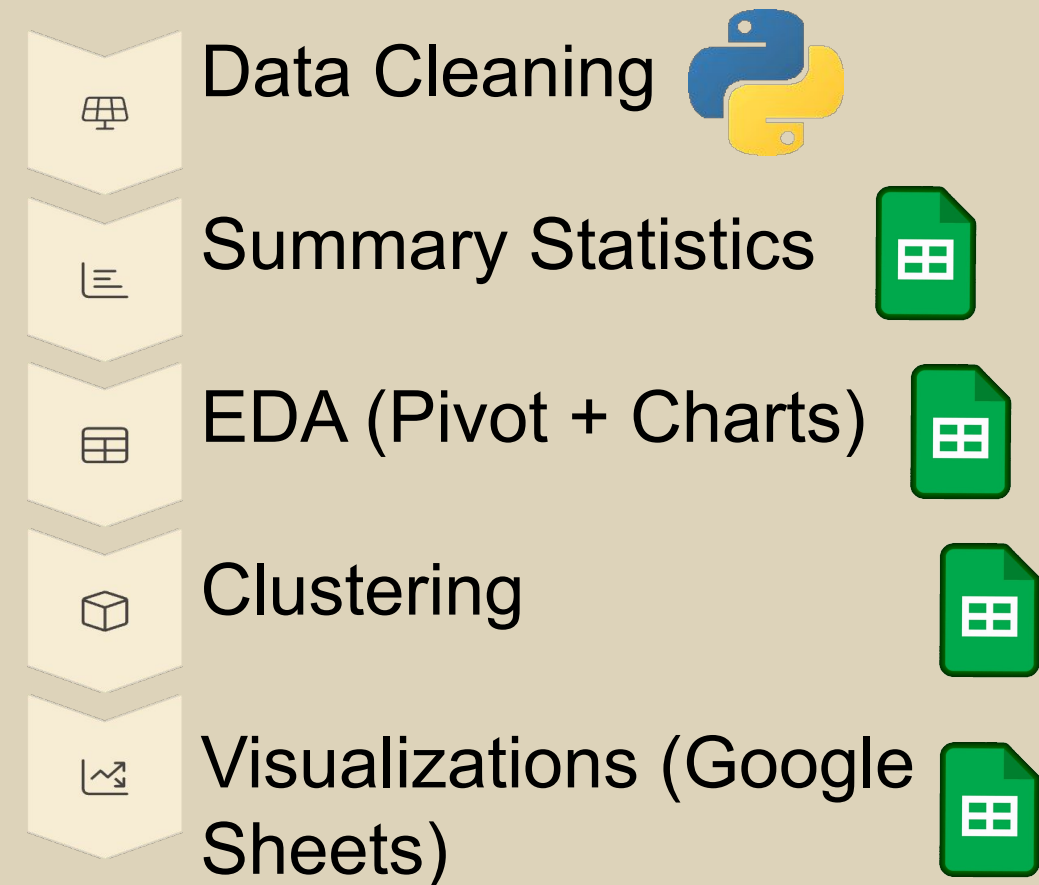
- Used pivot tables and charts in Google Sheets to analyze trends in orders, SKUs, fulfillment

Clustering & Segmentation:

- Grouped SKUs by sales, returns, category

Visualization Tools:

- Used Google Sheets for visuals
- Created treemaps, line charts, stacked columns, and Pareto charts



Key Findings – SKU & Order Analysis

1. Treemap: SKU Sales Distribution

- Shows how a few SKUs contribute to most of the revenue
- Highlights **zero or underperforming SKUs** (e.g., 19 SKUs with ₹0 sales)

SKU Sales Distribution

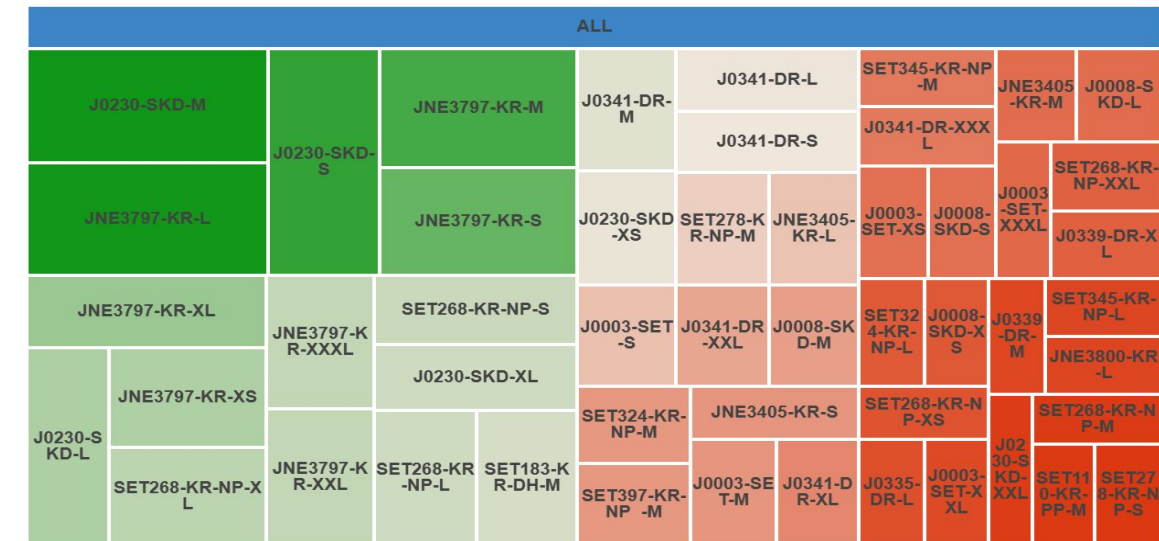


Figure 6.1: Contribution of SKUs to Total Revenue (many show zero or minimal sales)

2. Pie Chart: Order Status Breakdown

- Shows % of Delivered vs Cancelled vs Returned orders
- Reveals that ~10% of orders were either **cancelled or returned**

Order Status Distribution

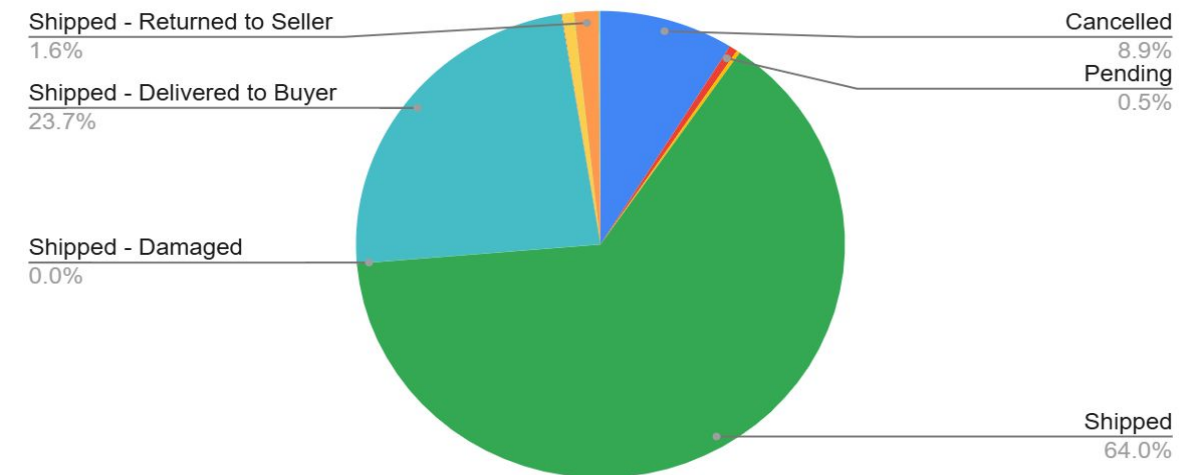


Figure 6.2: Proportion of Delivered vs Cancelled and Returned Orders

Key Findings – Fulfilment & Category Insights

Fulfilment Comparison:

- Amazon-fulfilled orders had better delivery success
- Merchant-fulfilled orders had **higher returns/cancellations**
- Recommends **prioritizing Amazon FBA** for efficiency and customer experience

Category Concentration:

- **Over 80%** of revenue comes from just **two categories**
- Risky in case of market saturation or demand shifts
- Recommends **exploring diversification into other fashion categories**

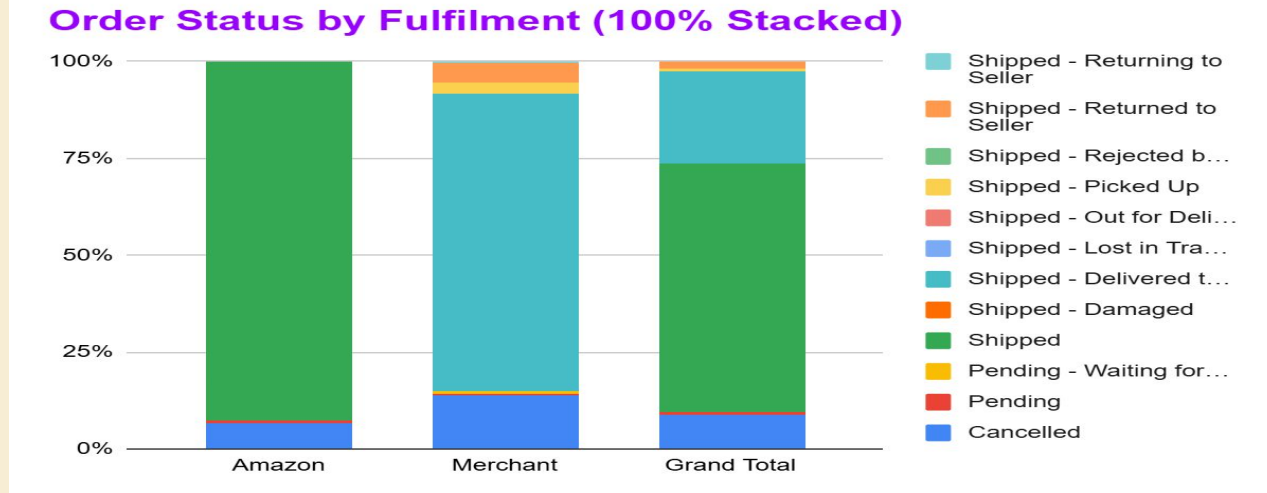


Figure 7.1: Order Fulfilment Performance – Amazon vs Merchant (Normalized %)

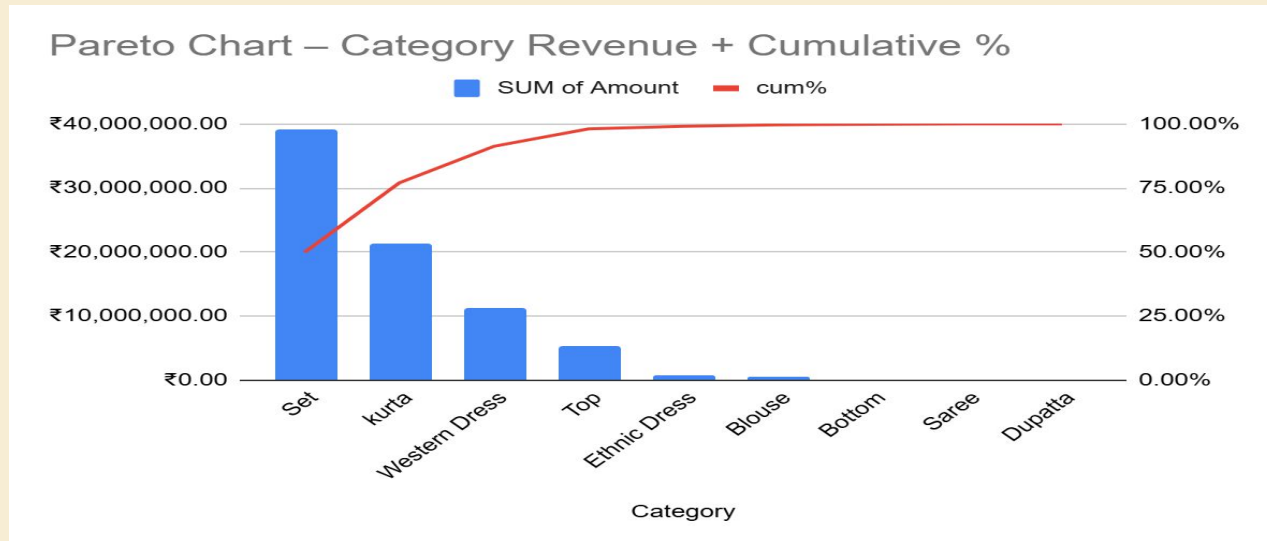


Figure 7.2: Pareto Chart – Dominance of Kurta and Set Categories in Total Sales

Key Findings – Region & Time Trends

Time Trend Insight:

- Daily revenue data shows **irregular patterns**
- Suggests demand surges may align with **weekends, festivals, or marketing efforts**
- Recommends identifying high-performance days for inventory alignment

Geographic Insight:

- **Top 5 states** (Karnataka, Maharashtra, Delhi, Tamil Nadu, etc.) dominate revenue
- Indicates these are **priority markets for promotions and fast shipping**
- Recommends regional warehouse or localized ads for optimization

Daily Sales Trend

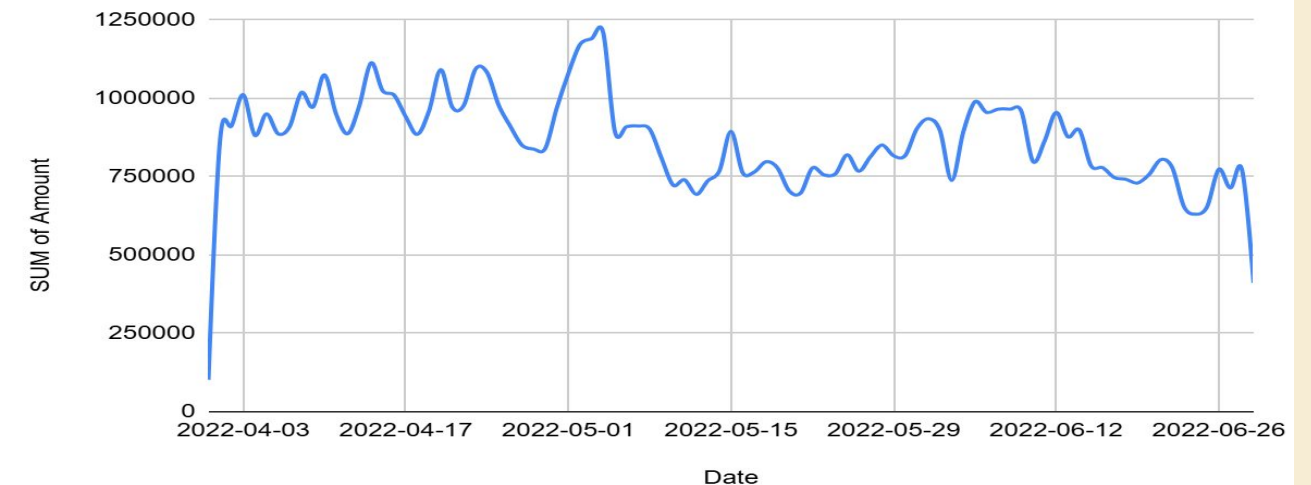


Figure 8.1: Line Chart Showing Daily Sales Trend – Peaks Indicate Potential Event-Driven Demand

Top 10 States by Sales

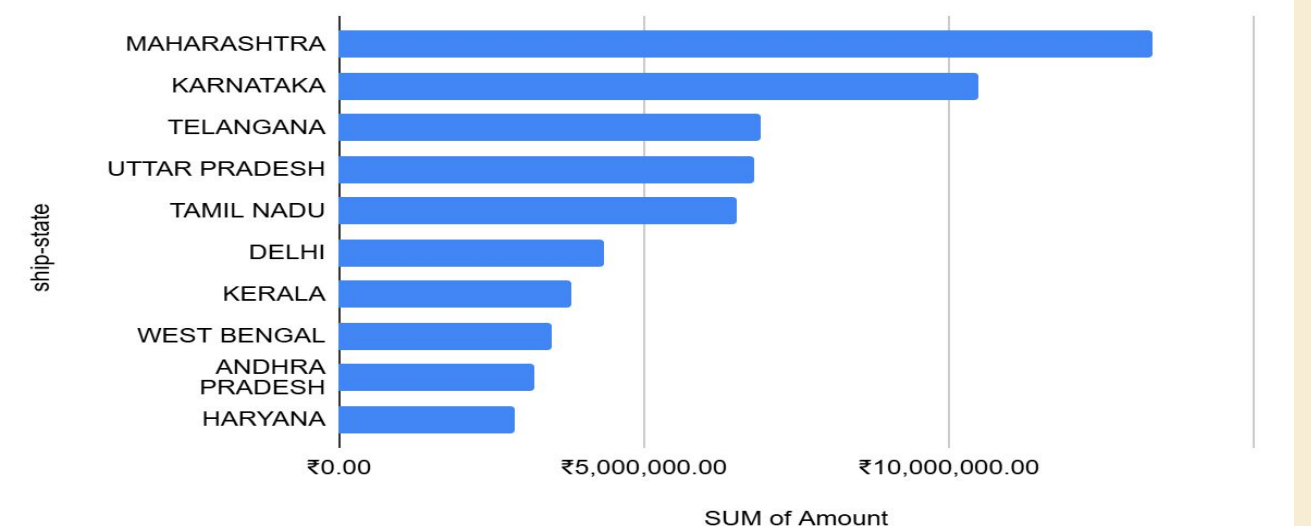


Figure 8.2: Revenue Contribution by State – Top Performing Regions

Recommendations

1 Optimize Inventory and SKU Strategy

- Eliminate or phase out **zero- and low-performing SKUs**
- Focus on **top-performing products** (Kurta, Set) but with diversification
- Run A/B tests on low-performing items to test listing improvements

2 Improve Fulfilment Approach

- **Prioritize Amazon Fulfilled (FBA)** to reduce returns and improve delivery experience
- Identify and correct issues in **Merchant-fulfilled order handling**

3 Address Return/Cancellation Drivers

- Enhance product listings: **accurate size guides**, clearer images, better descriptions
- Monitor categories with higher return rates and redesign listings

4 Expand Category Offerings

- Reduce **over-dependence on Kurta & Set**
- Explore western wear and fusion categories to reach broader audiences

5 Localize Marketing by Region

- Focus promotions in **high-revenue cities** (Bengaluru, Mumbai, Delhi)
- Consider **hyperlocal delivery** options in Tier-1 zones

Conclusion & Attachments

Conclusion:

The analysis identified key issues in SKU performance, fulfilment, and product concentration.

Dead inventory, high return rates, and category overdependence were major blockers to profitability.

By applying data-driven strategies, TrendyThreads Pvt. Ltd. can:

- **Streamline inventory**
- **Reduce cancellations**
- **Improve fulfilment efficiency**
- **Expand sustainable categories**

These actions will support **long-term profitability** and growth.

Attachments:

 [Cleaned Dataset \(.xlsx\)](#)

 [Python Notebook \(Google colab\)](#)

Thank you